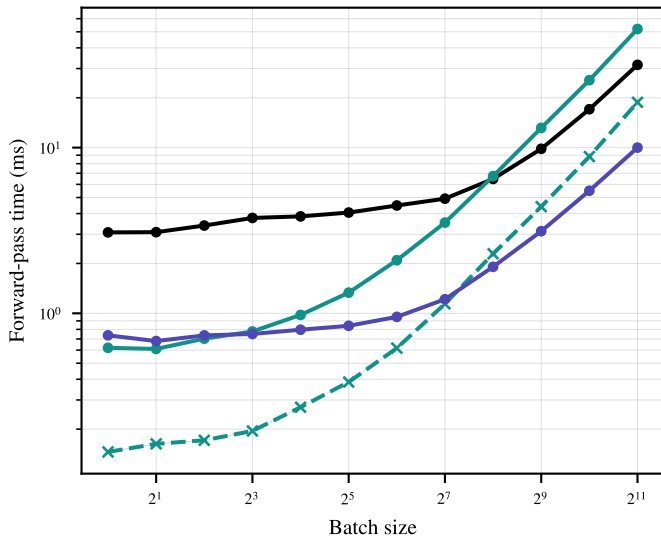


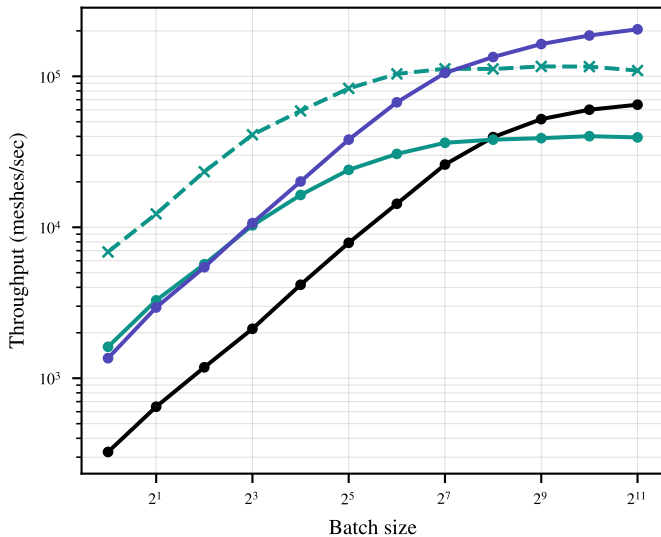
SOMA forward pass on RTX 5080

● SOMA-X (PyTorch + Warp)
 ● SOMA-JAX · full fit (SVD in XLA)
 ✕ SOMA-JAX · full fit (SVD in Warp)
 ● SOMA-JAX · linear fit (approx.)

Per-forward time (median, 95% CI < 1%)



Throughput (median, 95% CI < 1%)



mid-LOD SOMA mesh: 18056 verts, 78 joints · LBS-only · median over sustained-clock-warmed samples

skeleton fit: full = SOMA-X's exact RBF+Kabsch solve (rotation step = a 3x3 SVD) · linear = cheaper approximation (no SVD) · SVD in XLA / Warp = which kernel runs it